



Characterizing the influence of the marine stratocumulus cloud on the land fog at the Atacama Desert

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ABSTRACT

Fog in the Atacama Desert is a virtually untapped source of fresh water in the driest place on Earth. Focusing on understanding the role played by marine stratocumulus (Sc) in the development of land-fog, we analyse surface observations made along a steep transect at different heights. These observations are combined with numerical experiments performed using the Weather Research and Forecasting Model (WRF). We find two main diurnal regimes based on atmospheric thermal stability, both of which determine the formation and dissipation of fog. These are (a) a well-mixed regime characterized by low gradients of potential temperature and specific humidity, low diurnal variability, and presence of Sc cloud-fog. (b) A stratified regime characterized by high gradients of potential temperature and specific humidity, high diurnal variability, but no Sc clouds nor presence of fog. By using the parcel method, initialised with surface observations, we characterize the Sc cloud of nine typical fog events, estimating a mean cloud depth of 566 m between 740 m (± 150 m) and 1307 m (± 30 m). Fog observations at ground level agree with these cloud-base and cloud-top estimates, showing a liquid water mixing ratio of Sc cloud-fog in the range $0.3\text{--}0.7\text{ g kg}^{-1}$. The study reveals that the advection of marine Sc cloud and the stability of the boundary layer are key processes in the formation and dissipation of fog. Sc cloud advection over land is modulated by upwind and driven by topography and local circulation. We conclude that a realistic characterization of Sc cloud-fog is possible by combining limited surface observations and numerical experiments.

1. Introduction

Fog in arid regions is regarded as one of the few viable freshwater resources, playing a key role in ameliorating water scarcity (Klemm et al., 2012). In the Atacama Desert, water scarcity produces tensions between industry, civil society, and the environment, all of which are dependent on the limited water resources available to develop (Oyarzún and Oyarzún, 2011). Water is needed for human consumption, but also for energy and food production, the fundamental basis of development (Schewe et al., 2014). The high hydrological potential of the fog harvesting in the Atacama Desert is due both to the interaction between the marine Sc cloud with the abrupt orography, and to its proximity to major population centres such as Antofagasta, Arica and Iquique (Schemenauer and Cereceda, 1994b). Due to these two characteristics, the fog is one of the most viable freshwater resources to be tapped. The Atacama Desert is believed to be the driest place on Earth, characterized by rainfall of 0.1 mm per decade and large daily temperature variations between 5°C and 30°C (Weischet, 1975). However,

the high humidity at the coast provided by the proximity of the Pacific Ocean contrasts with the extreme aridity of the desert itself (Cereceda et al., 1991 and Cereceda et al., 2000). Like other deserts, the Atacama Desert lies on the west face of the continent and owes its extreme aridity to three main processes (Weischet, 1975): (1) downward vertical air masses leading to subsidence motions caused by the Hadley cell, (2) cold-water upwelling led by the Humboldt current, and (3) the steep topography of both the Andes and the coastal mountain ranges, which prevent the humidity from being transported from the Amazon and the Pacific Ocean. Above the ocean, the first two processes also enhance the formation of a Sc cloud deck that covers a large portion of the Sub-tropical Southeast Pacific (SSEP) around $14^\circ\text{--}30^\circ\text{S}$ (Cereceda et al., 1999). These clouds are transported by the wind until they collide with the steep topography of the mountain ranges (Fig. 1). The coast of the Atacama Desert is characterized by high humidity and low thermal oscillations (Cereceda et al., 2008a). Due to the humidity input from the Pacific Ocean, fog persists between 650 m and 1200 m above sea level (a.s.l.) (Cereceda et al., 2002). The coastal region of the desert lies

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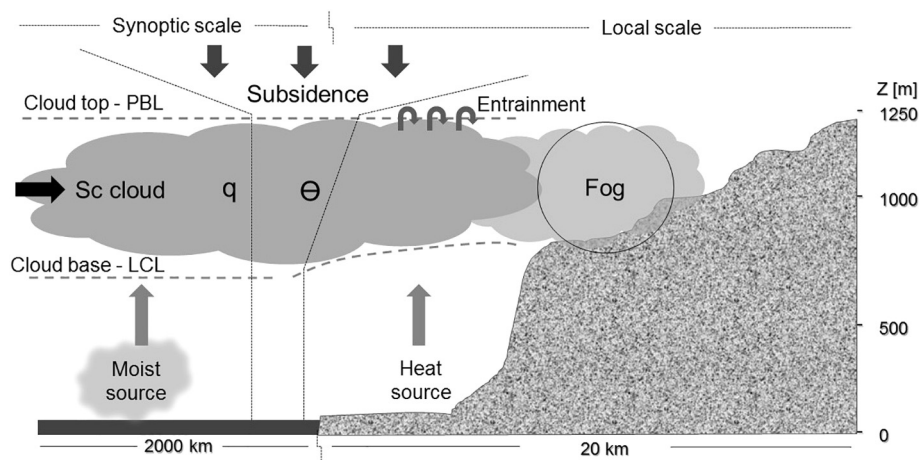


Fig. 1. Physical processes associated with marine Sc clouds and the land-fog system (MS-LF).

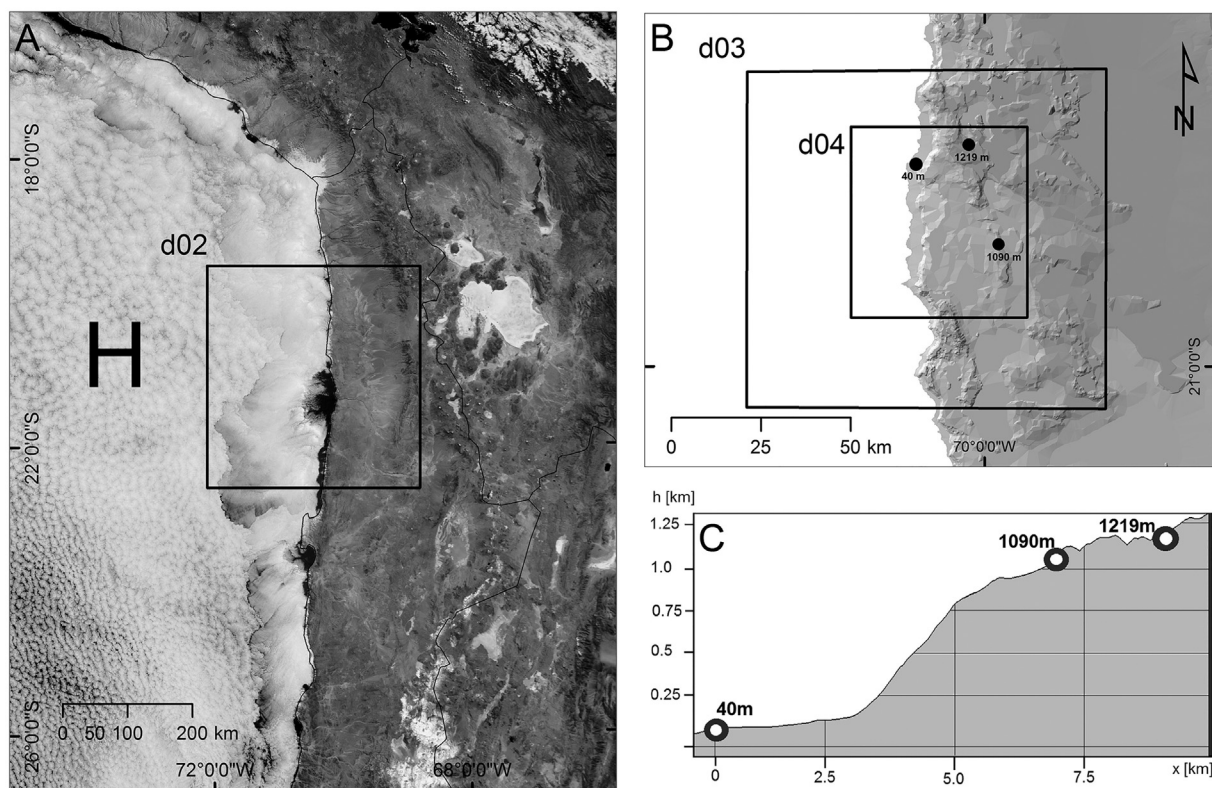


Fig. 2. Research area; (a) Atacama Desert location MODIS image on July 21st, 2015 and outer WRF model domain. (b) Spatial distribution of surface observations and inner WRF model domains. (c) Schematic cross-section of the vertical distribution of surface observations.

between 18.5° and 23.5° S, around 70° W and within 20 km of the shore (Fig. 2). The fog has been largely studied in terms of climate, ecosystems, and water production. del Río et al. (2018), describe the inter-annual fog variability and its relationship with macroclimatic phenomena. The spatial and temporal distribution of Sc clouds (Osses et al., 2005; Farías et al., 2005 and Cereceda et al., 2008b) and the relict vegetation in the so-called fog oases (Cereceda et al., 1991; Cereceda et al., 2000 and Muñoz-Schick et al., 2001) has also been widely described. Regarding the potential freshwater uses, by using standard fog collectors (Schemenauer and Cereceda, 1994a) researchers have measured annual means of fog collected between 1.1 and 7 L m^{-2} collector per day in the range of 850 m and 1100 m a.s.l. (Cereceda et al., 2008b). However, the quantity and quality of their observations were limited (Cereceda et al., 2008a), mainly due to the lack of meteorological

stations and the low frequency of the measurements. For this reason, we still lack an in-depth understanding of the processes that control fog formation, maintenance and dissipation (Cereceda et al., 2008a), particularly with respect to the physical characterization and quantification of the fog and its relation with the marine Sc cloud. Here, we take a fresh look at the fog phenomenon, from a physical and meteorological perspective. By combining surface observations collected at three different meteorological stations in a transect and regional modeling results, we aim to answer the question: what is the physical influence of the marine Sc cloud in the formation, maintenance and dissipation of the land-fog in the Atacama Desert? The combination of different sources of surface observations with regional modeling results aims for first time to compensate for the lack of high-quality measurements. The surface observations, which included meteorological stations, satellite

images and fog collection measurements, were analysed together with the WRF model. This combination of sources enabled us to characterize the vertical and horizontal scales of the marine Sc cloud and its influence on the land-fog system (MS-LF) in their seasonal and diurnal variability.

2. Key physical processes under study

The fog in the Atacama Desert is a complex system of interactions between atmosphere and land that range from local to synoptic length scales. The marine Sc cloud deck formed at SSEP covers length scales of 1000 km from 14° to 30° S. However, the fog produced by the Sc cloud at land just occurs on a length scale of 10 km or less. Our view of the processes of the MS-LF in the Atacama Desert is sketched in Fig. 1.

The marine Sc cloud deck is a synoptic phenomenon formed as a result of the interaction of two main fluxes over the ocean; surface heat fluxes and subsidence. The left side of Fig. 1 depicts the synoptic scale processes. The upward heat and moisture fluxes move from the ocean surface to the top of the atmospheric boundary layer (ABL), where they mix well, condense and form clouds (Bretherton and Park, 2009). This wet and well-mixed air parcel is blocked by the inversion capping caused by the downward flux of dry air (Weischet, 1975), which in turn is due to the subsidence that the Hadley cell produces in subtropical latitudes (Chang, 1995). Here, as a result of the instability created by the longwave radiative cooling at the cloud top, dry air entrains into the cloud layer, producing an exchange between the ABL and the free troposphere (Duynerkerke et al., 1995). The Sc cloud deck is transported by the horizontal wind from the south-southwest (S-SW) towards the coastal mountain range of the Atacama Desert. This phenomenon takes place mainly between 18° and 26° S (Cereceda et al., 2000 and Cereceda et al., 2008a). The horizontal wind transports the Sc cloud deck towards the land where, we hypothesize, it is affected by the buoyancy produced by the land surface heat fluxes that forces the cloud base to rise, leading to the thinning of the cloud deck. We also suggest that the high and steep topography of the coastal Atacama Desert produces an uplifting of the cloud layer, which covers the land until it gradually dissipates inland. These processes occur on a reduced length scale of between 10 km and 50 km. The contact between the Sc cloud deck and the topography is known as fog in the Atacama Desert (Cereceda et al., 2000) (Fig. 1). Cereceda et al. (2002), described a general classification for this fog in the Atacama Desert, based on its origin; orographic, advective and radiation fog. This classification is based on its spatial distribution, its effect on the ecosystems and the amount of fog collected. The two first characteristics are formed in the coastal region of the Atacama Desert. The orographic fog is produced by the humid air parcel transported by the wind, which rises due to the presence of the high topography, condensing adiabatically and thus leading to fog formation (Cereceda et al., 2000). The advective fog is produced by the Sc cloud deck above the ocean once it arrives on land (Cereceda et al., 2008b). When it encounters the orography, clouds at ground level are formed, thus the fog (Klemm et al., 2012). We focus our research on the advective fog, which has been reported by Cereceda et al. (2002) and Cereceda et al. (2008b) as the most frequently observed fog in the research area. We then propose a new classification, based on the stability of the ABL, which takes potential temperature and specific humidity as classification criteria.

3. Methods

3.1. Surface observations

Surface observations of air temperature, pressure, relative humidity, wind speed and wind direction and fog collection were analysed to characterize the evolution of the ABL. The observations made at a resolution of one hour were collected from three meteorological stations located at 40 m a.s.l., 1090 m a.s.l. and 1219 m a.s.l., and fog

observations from stations at 1090 m a.s.l. and 1219 m a.s.l. within 20 km of the coast. The fog measurements are taken by a standard fog collector (SFC), which consists of a frame of 1.0 m² of double polypropylene raschel mesh with a 35% shading coefficient (Schemenauer and Cereceda, 1994a). The units are liters of water per square metre (L m⁻²), which is equivalent to a depth of water of 1 mm over 1 m² of the collector surface. The time unit depends on the data resolution; in our research we use an hourly resolution. Fig. 2a shows the contrast between the sea, characterized by a Sc cloud deck, and the land, characterized by a steep topography where fog develops intermittently near the coastline. Fig. 2b shows the spatial distribution of the meteorological stations and Fig. 2c a schematic cross-section of their vertical distribution. The meteorological station located at 40 m a.s.l. belongs to the Chilean Civil Aviation Authority (DGAC), at Diego Aracena airport. The stations at 1090 m a.s.l. and 1219 m a.s.l. belong to the Pontifical Catholic University of Chile and Heidelberg University respectively. We focus on the data analysis in 2015, since most of the stations were continuously collecting data starting that year.

Nine representative days of fog events were selected for autumn, winter and spring on the basis of fog collection, seasonality and the quantity and quality of the observations. We calculated the potential temperature (θ) and specific humidity (q), using pressure observed at 40 m a.s.l. as a reference. This means that we calculate the changes of a parcel at 1090 and 1219 m a.s.l. which is forced to descend to 40 m a.s.l. (p_0). This enabled us to determine whether the air masses at 40 m have similar characteristics to those of the air masses at 1090 m and 1219 m. On the basis of these selection criteria, we determined the difference in the conserved variables between the highest level (1219 m) and the reference level (40 m). Then, we define the following regimes for the nine cases (days) under study:

$$\frac{\partial \theta}{\partial z} \approx \frac{\Delta \theta}{\Delta z} \begin{cases} < 3.10 \cdot 10^{-3} \text{ K m}^{-1} & \text{Well - mixed} \\ > 3.10 \cdot 10^{-3} \text{ K m}^{-1} & \text{Stratified} \end{cases} \quad (1)$$

$$\frac{\partial q}{\partial z} \approx \frac{\Delta q}{\Delta z} \begin{cases} < 1.60 \cdot 10^{-3} \text{ g kg}^{-1} \text{ m}^{-1} & \text{Well - mixed} \\ > 1.60 \cdot 10^{-3} \text{ g kg}^{-1} \text{ m}^{-1} & \text{Stratified} \end{cases} \quad (2)$$

3.2. Cloud characteristics

The cloud characteristics are determined by estimating the cloud base (CB) and the liquid water mixing ratio (q_l) of the cloud by the parcel method, and the cloud top (CT) through a combination of remote-sensing procedures. The cloud characteristics were inferred only for the well-mixed regime days because the θ is conserved until the lifting condensation level (LCL) and q is conserved along the ABL.

To determine the CB, we utilized the parcel method (Wetzel, 1990), which assumes that an air parcel is lifted from the surface without mixing during the ascension following the dry adiabatic until the LCL. The parcel method enables us to infer the vertical profile of ABL in terms of potential temperature (θ), specific humidity (q) and adiabatic water liquid mixing ratio (q_{la}) via surface observations. These thermodynamic variables enable us to calculate the height at which the air condenses (LCL), where specific humidity equals the saturated specific humidity ($q = q_{sat}$). As Sc clouds form within the ABL, we assume the LCL to be the same as the CB. In our observations surface pressure at 40 m a.s.l. is used instead of 1000 hPa as a pressure reference level for all calculations, employing the parcel method with this reference level for 1090 m, and 1219 m. Thus, we did the following: (a) we calculated the changes in an air parcel ascending from 40 m by using θ , p and q of the station at 40 m a.s.l. and (b) we calculated the changes in an air parcel ascending from 40 m by using θ and q of the stations at 1090 and 1219 m a.s.l.

The q_l was estimated by using simple criteria as $q_l = q_s - q$, if $q_s \geq q$, obtaining a first rough estimate. In order to obtain more realistic values

of q_i we developed a simple parameter m to take into account the mixing of air into the parcel method. This parameter is inspired by the method used by [Wetzel and Boone \(1995\)](#). The mixing rate parameter m , which ranges from 0 to 1, represents the amount of air in the environment that is entrained into the parcel along with the rise of the parcel from the surface to the top of the mixed layer. For $m = 0$, there is no air entrained and the parcel conserves its original θ from the surface. For $m = 1$, the air in the parcel is identical to that of the mixed layer when it reaches the top of the mixed layer. We assume that m does not change throughout the mixed layer, so that the contribution of entrained air to the parcel properties for $m \neq 0$ is linear with height. We also assume that the mixing rate is independent of time, height and properties of parcel and environment and is identical for temperature and moisture. Using these assumptions and the variables presented, the potential temperature and specific humidity of the parcel at a height z read:

$$\theta^p(z) = \left(1 - m \frac{z}{z_{LCL}}\right) \theta^s + m \frac{z}{z_{LCL}} \theta^{ML} \quad (3)$$

$$q^p(z) = \left(1 - m \frac{z}{z_{LCL}}\right) q^s + m \frac{z}{z_{LCL}} q^{ML} \quad (4)$$

The variables representing the state of the parcel are potential temperature θ^p and specific humidity q^p . Additional variables used by the parameterization are the height of the parcel z , the height of the lifting condensation level z_{LCL} , the liquid potential temperature and total specific humidity of the mixed layer θ^{ML} , q^{ML} and at the surface θ^s , q^s , and the (constant) mixing rate parameter m .

Finally, the cloud top (CT) is determined by combining the MODIS cloud cover fraction (CC) and Aster-GDEM digital elevation model. We calculated the mean elevation where the Sc cloud deck collides with the coastal cliffs. We base our calculations on two criteria. The first is that 100% CC needs to overlap a steep slope ($> 25^\circ$), because over flat slopes the cloud can be several metres over the land surface and does not necessarily touch it. The second criterion is that mean elevation must be above 1219 m because at this altitude we have the highest fog observations. The values of CT obtained correspond to one measure per day and this study assumes them to be constant throughout the day.

3.3. Model setup

The observational analysis is complemented by a numerical experiment performed by the Weather Research and Forecasting Model (WRF) to present the Sc cloud influencing the land. The days selected for the modeling were July 21st and 28th, 2015, which were characterized by well-mixed and stratified conditions, respectively. We used initial boundary-layer conditions of ECMWF for $-21/-70$ (lat/long) with a time resolution of six hours. [Fig. 2a](#) and [b](#) show the horizontal resolution and domains. Sixty levels are set, including 42 in the first 2000 m. We corrected the sea surface temperature (SST) by $+1.5$ K in accordance with NOAA observations. The main parameterizations utilized were; for boundary layer physics, the UW boundary layer scheme from CAM5 proposed by [Bretherton and Park \(2009\)](#) and for convection physics the modified Tiedtke scheme proposed by [Zhang et al. \(2011\)](#). The complete model setup can be found in Appendix 1.

4. Atmospheric boundary layer regimes

We identify two regimes, based on the analysis of nine fog events observed in 2015 at MS-FL zone. The first regime characterizes six days under a well-mixed thermal and moist layer. The second regime, describes three days under a stratified thermal and moist layer condition. These regimes are strongly associated with the formation, maintenance and dissipation of Sc cloud-fog.

[Table 1](#) summarises the classification and quantification of surface

observations for both ABL regimes. It first classifies the regimes according to date and seasonality and then quantifies the stability parameters between the lowest and highest observational stations (40 m - 1219 m) as the gradient of potential temperature ($\partial\theta/\partial z$), the same gradient, but filtering the diurnal variability ($\partial\bar{\theta}/\partial z$ no diurnal) and the gradient of specific humidity ($\partial q/\partial z$). [Table 1](#) also includes measurements of wind speed and direction measured at 10 m at 40 m a.s.l. (ws-40, wd-40), total fog collected (fog_c) and hours of thermal diurnal variability (diurnal-var). Finally, calculations of the cloud base (CB) and cloud top (CT) heights, obtained via the parcel method, are shown.

The well-mixed regime differs from the stratified regime by having low differences in the diurnal $\partial\bar{\theta}/\partial z$ and the non-diurnal $\partial\bar{\theta}/\partial z$ ([Table 1](#)). For the well-mixed regime, the diurnal variability time of the θ is shorter than 12 h (Diurnal-var). Likewise, the $(\partial q/\partial z)$ in well-mixed regimes is lower than in the stratified regimes, exhibiting peaks below $1 \times 10^{-4} \text{ g kg}^{-1} \text{ m}^{-1}$. The mean daily wind comes from S-SW for the nine days analysed, which shows that the marine Sc cloud moves inland under both regimes. They do not display differences in wind speed between regimes, but the wind-speed is higher in spring than in autumn-winter ([Table 1](#)). The days analysed do not display important differences in the amount of fog collected. Both wind direction and fog collected values are similar to data reported by [Cereceda et al. \(2008a\)](#), for the same seasons (SW winds and 7 L m^{-2} per day). Finally, the CB and CT are shown only for days dominated by a well-mixed regime, presenting a cloud depth average of 566 m.

In order to characterize both regimes in greater depth, we now focus on two representative fog events. [Fig. 3a](#) and [b](#) show the diurnal observations of θ and q at heights 40 m, 1090 m and 1219 m, on July 21st, 2015 as a representative example of the well-mixed regime. Likewise, [Fig. 4a](#) and [b](#) show the same diurnal observations, but for a typical stratified regime on July 28th, 2015.

4.1. Well-mixed regime

The well-mixed regime is characterized mainly by low gradients of potential temperature and moisture within the ABL ([Fig. 3a](#) and [b](#)). The daily evolution of θ shows a collapse of the three curves of the observational stations during the night-morning and evening, 0:00 to 12:00 and 18:00 to 00:00 local time (LT). The θ between the lowest and highest station differs by $< 2 \text{ K}$ (daily gradient between $1.84 \cdot 10^{-3}$ and $2.36 \cdot 10^{-3} \text{ K m}^{-1}$; [Table 1](#)). Under these conditions, we assume that conserved variables are independent of height, following the well-mixed conditions of the marine Sc cloud. The well-mixed properties are only interrupted by the diurnal variability of inland measurements (1090 m and 1219 m) between 12:00 and 18:00 LT.

The q evidences almost no diurnal variation for the three stations ([Fig. 3b](#)). It also shows a small daily mean difference in moisture of 0.52 g kg^{-1} in the whole ABL height (moisture gradient of $4.41 \cdot 10^{-4} \text{ g kg}^{-1} \text{ m}^{-1}$, [Table 1](#)). The specific humidity is not affected by diurnal variability according to our criteria, which shows that a mixing of the entire ABL takes place throughout the day. The fog was observed in the afternoon, evening and night on July 21st, 2015, including peaks above 2 L m^{-2} and a total amount of 13.4 L m^{-2} . Collected fog-water on the six days classified as well-mixed regimes were similar to those reported by [Cereceda et al. \(2008a\)](#) (annual means of 7 L m^{-2} per day). Fog collection was always observed when the ABL is well-mixed in θ and q , for example, before 12:00 and after 18:00 LT in [Fig. 3a](#). Related to the regimes described, daily mean wind direction is S-SW of 249°N , which shows the horizontal transport of oceanic air masses inland ([Fig. 2a](#) and [Fig. 10](#)). Wind-speeds ranged around 6 m s^{-1} and were mostly stable under this regime.

Marine air masses thus influence the land fog system under well-mixed conditions. This influence is clear in the observations made close to the ocean at 40 m a.s.l., which has a lower diurnal variation than observations made inland, for example at 1090 m and 1219 m ([Fig. 3a](#) and [b](#)). This has two main reasons. First, observations were made in a

Table 1
Surface observations and inferred Sc cloud characterization for the well-mixed (WM) and stratified (ST) regimes. The regimes are described by the dates of the observations (Day); season autumn (A), winter (W) and spring (S); the potential temperature gradient between the highest and lowest stations ($\partial\bar{\theta}/\partial z$); the potential temperature gradient after filtering out the diurnal variation ($\partial\bar{\theta}/\partial z$ no diurnal); the specific humidity gradients between the highest and the lowest stations ($\partial\bar{q}/\partial z$); wind speed and direction measured at 10 m at 40 m a.s.l.; total fog collection per day (fog_t); hours of diurnal variability; cloud base (CB) and cloud top (CT).

Day	Regime	Season	$\partial\bar{\theta}/\partial z$	$\partial\bar{\theta}/\partial z$ non – diurnal	$\partial\bar{q}/\partial z$	ws – 40	wd – 40	fog _t	Diurnal – var	CB	CT
			[K m ^{−1}]	[K m ^{−1}]	[g kg ^{−1} m ^{−1}]	[m s ^{−1}]	[°N]	[L m ^{−2}]	[hrs]	[m]	[m]
18–04–15	WM	A	$3.45 \cdot 10^{-3}$	$2.02 \cdot 10^{-3}$	$1.78 \cdot 10^{-4}$	5.63	211	7.3	9.5	886.3	1285
29–04–15	WM	A	$5.27 \cdot 10^{-3}$	$3.10 \cdot 10^{-3}$	$4.58 \cdot 10^{-4}$	6.17	166	1.1	12	860.5	1273
21–07–15	WM	W	$2.36 \cdot 10^{-3}$	$1.84 \cdot 10^{-3}$	$4.41 \cdot 10^{-4}$	6.29	249	13.5	6	784.1	1366
08–08–15	WM	W	$2.41 \cdot 10^{-3}$	$1.87 \cdot 10^{-3}$	$9.75 \cdot 10^{-4}$	6.83	199	4.1	7	651.5	1308
17–09–15	WM	W	$3.21 \cdot 10^{-3}$	$1.68 \cdot 10^{-3}$	$7.72 \cdot 10^{-4}$	9.23	202	7.1	10	667.9	1320
08–10–15	WM	S	$3.22 \cdot 10^{-3}$	$2.18 \cdot 10^{-3}$	$7.89 \cdot 10^{-4}$	9.46	202	6.6	10	593.7	1292
27–07–15	ST	W	$1.21 \cdot 10^{-2}$	$1.09 \cdot 10^{-2}$	$3.45 \cdot 10^{-3}$	6.79	180	5.6	20	–	–
28–07–15	ST	W	$9.74 \cdot 10^{-3}$	$6.19 \cdot 10^{-3}$	$2.82 \cdot 10^{-3}$	5.29	212	7.4	14	–	–
20–10–15	ST	S	$1.08 \cdot 10^{-2}$	$9.41 \cdot 10^{-3}$	$1.65 \cdot 10^{-3}$	9.25	188	–	20	–	–

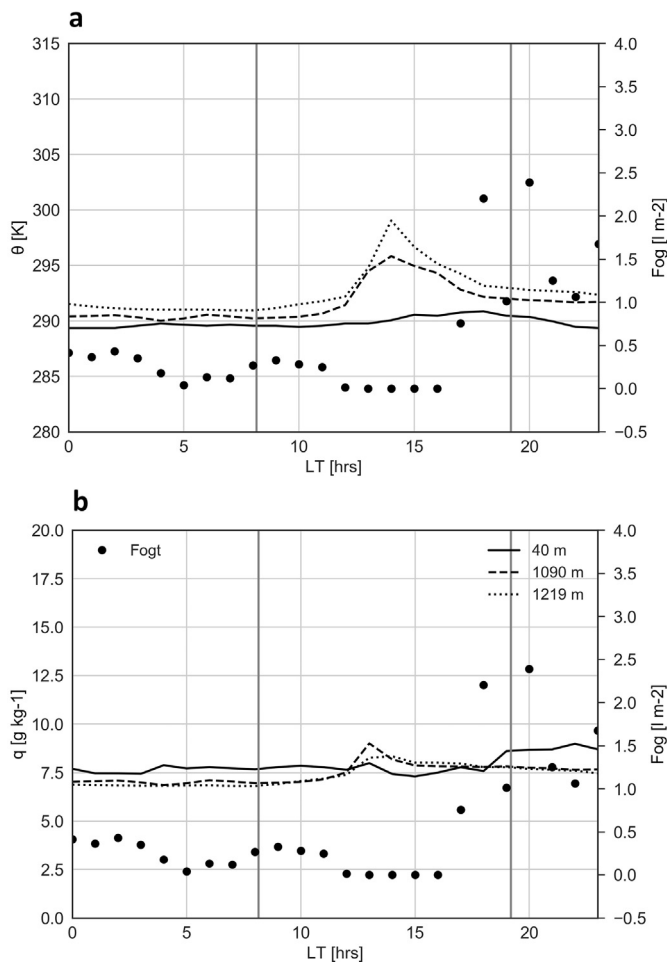


Fig. 3. Hourly average for (a) θ and (b) q for the well-mixed day of July 21st, 2015. Values represented for stations located at 40 m (solid line), 1090 m (dashed line), and 1219 m (dotted line). The sum of fog collected at these three levels (fog_t) is represented by black dots. Sunrise and sunset are shown by grey vertical lines.

transition zone (MS-LF), which lies between the marine boundary layer (MBL), which is characterized by the Sc cloud presence, and the continental boundary layer (CBL), characterized by arid conditions. Secondly, as Sc cloud drives under well-mixed conditions above the ocean (Duynderke et al., 1995), we relate well-mixed regimes to the advective fog phenomenon reported by Cereceda et al. (2002). However, our

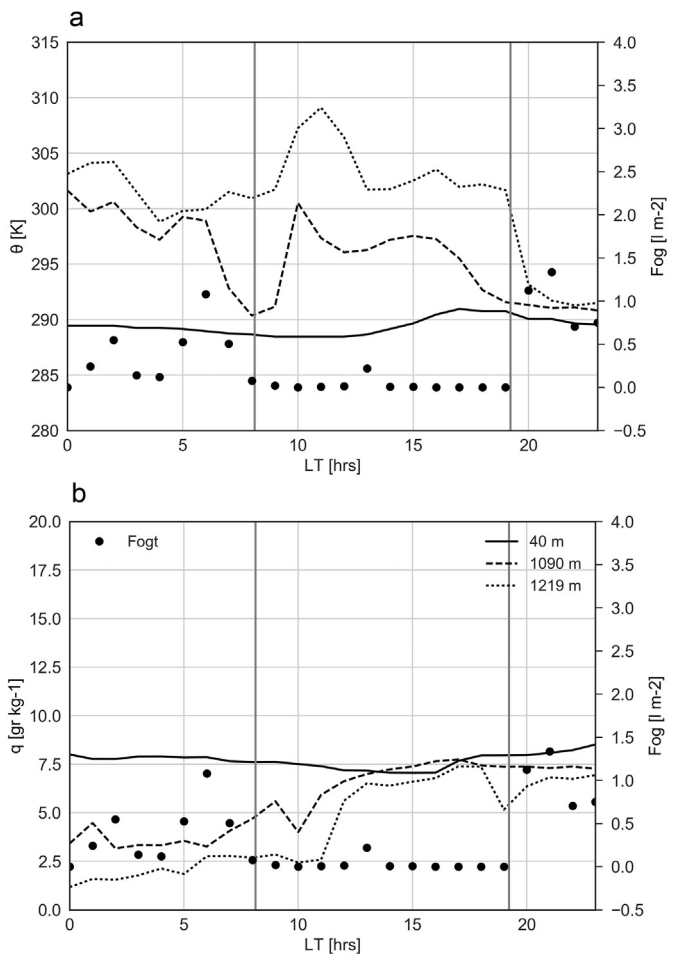


Fig. 4. Hourly average for (a) θ and (b) q for the stratified day July 28st, 2015. Values for stations placed at 40 m (solid line), 1090 m (dashed line), and 1219 m (dotted line). Sunrise and sunset are shown by grey vertical lines. The sum of fog collected at these three levels (fog_t) is represented by black dots. Sunrise and sunset are shown by grey vertical lines.

findings also demonstrate that well-mixed regimes are related to orographic fog (Cereceda et al., 2002). The latter is due to the collapse of the curves shown in Fig. 3a and b, which suggests lifting of the same air parcel as it faced a steep slope. In this case, the air parcel contains the same θ and q at three levels. These findings are further corroborated by the Sc cloud characterization and the WRF numerical experiment.

4.2. Stratified regime

Unlike well-mixed regimes, the stratified regime is characterized by large gradients in potential temperature and moisture. As Fig. 4a shows, $\partial\bar{\theta}/\partial z$ reaches $9.74 \cdot 10^{-3} \text{ K m}^{-1}$ (Table 1) around midday. This means a difference in 11.5 K between the lowest and highest observational station. In common with most marine places, the station at 40 m a.s.l. does not show a diurnal variation, but the stations at 1090 m and 1219 m display a large diurnal variability reaching the highest θ of 310 K around midday.

This regime also exhibits large differences in moisture within the ABL (Fig. 4b). The station at 40 m shows the highest q of around 9 g kg^{-1} from 00:00 to 11:00 LT, while the stations at 1090 m and 1219 m q decrease at a rate of $4.2 \cdot 10^{-3} \text{ g kg}^{-1} \text{ m}^{-1}$. However, from midday (12:00 LT) onwards, q is similar at all three observational stations. Fog collection is observed mainly during the night and evening (00:00 to 09:00 LT and from 20:00 LT). These diurnal measurements are similar to the fog collected under days dominated by well-mixed regime, but with lower peaks. The wind direction blows from SW (212°N), which shows irrespective of the regime involved, oceanic air masses are transported inland (Fig. 10). Wind-speed increases under stratified regimes from 6 to 10 m s^{-1} .

The stratified conditions are explained by the CBL's stronger control of the MS-LF. On the one hand, the measurements at 40 m reveal typical marine conditions that vary only slightly, while those made at 1090 m and 1219 m, which additionally lie 20 km inland, display greater variations in θ and q at night and morning. This large variation is evidence of a typical nocturnal stable layer (Stull, 1988), which is stratified in both temperature and moisture. The latter process is enhanced by the radiative cooling that takes place at night, due to the increases in subsidence and katabatic winds (Fig. 10a). Thus, typical desert conditions (CBL) influence the MS-LF more than the MBL does during the night, which explains the stratification and dissipation of clouds and fog.

Even during a day dominated by stratified conditions, which dissipates clouds, fog could still be observed at night between 00:00 and 06:00 LT (Fig. 4). We have two hypotheses to explain this: the first is that when fog forms at high levels (1090 m and 1219 m), there are small differences in θ and q around 06:00 LT. However, they still present large differences with respect to the lowest level (40 m) (Fig. 4a and b). This means that only the highest layer of the ABL is well-mixed. We hypothesize that under these conditions, a thin cloud is advected by the wind inland (Cereceda et al., 2002), but it is decoupled from the bottom of the ABL. Our second hypothesis is that the residual turbulence of the previous day enhances mixing during the night at the top of the ABL. Thus, after the convection mixes the ABL, a residual layer might have continued to be turbulent at night (Stull, 1988). This could keep the ABL well-mixed and enhance Sc cloud formation at MS-LF, leading to fog on land. To summarise: on the one hand, the well-mixed conditions are closely related to the advection of marine Sc clouds over the land, which leads to the formation and maintenance of fog over the land. On the other hand, under stratified conditions, Sc clouds and fog dissipate. Even though the regimes suggested characterize an entire day, both regimes might occur in the course of a day. For example, we observe a stratified regime around 15:00 LT on July 21st, 2015 day that is dominated by a well-mixed regime. Likewise, we observe a well-mixed regime after 20:00 LT on July 28th, 2015 day that is dominated by a stratified. The regimes are further studied in the following sections by means of the parcel method of analysis and the WRF numerical experiments.

5. MS-LF physical characterization

This section describes the cloud characteristics of well-mixed regimes in terms of cloud base (CB), cloud top (CT) and liquid water mixing ratio (q_l). We used surface observations made at the steep slope

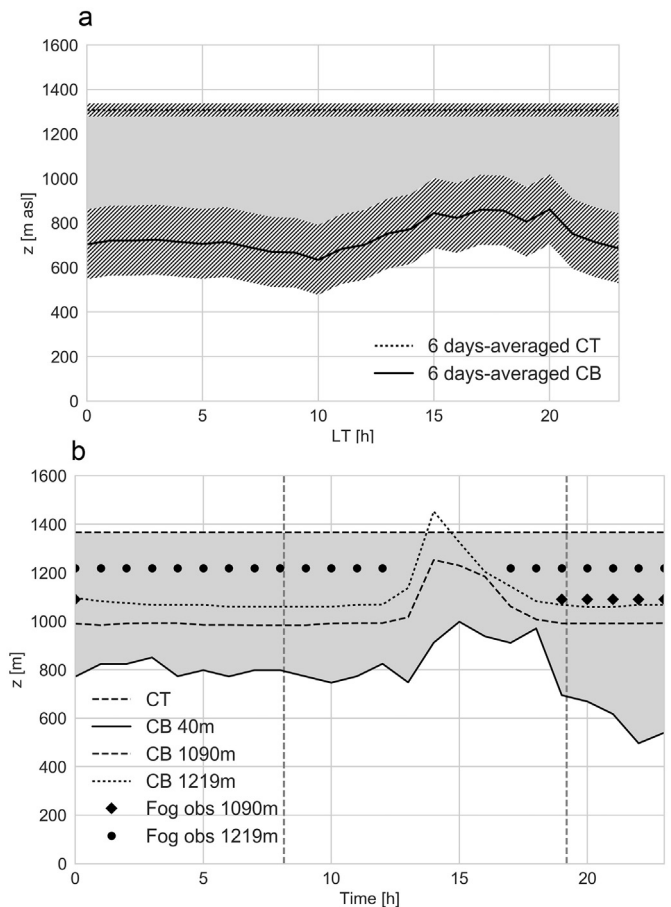


Fig. 5. (a) Six-day averaged cloud depth, calculated from the reference station (40 m) for 2015. (b) Diurnal variability of cloud depth over surface observations at 40 m, 1090 m and 1219 m, calculated from the reference station (40 m) for July 21st, 2015. Fog observation altitude is represented by dots and diamonds.

to validate the parcel method. These found no significant differences between the two approaches ($p \leq .05$). The following sections describe our findings regarding MS-LF characteristics, by means of an analysis of our surface observations.

5.1. Sc cloud-fog depth characteristics

Six days with a well-mixed regime in autumn, winter and spring were analysed (Table 1). Fig. 5a shows a six-day-averaged CB of 740 m with a variability of 150 m, a CT of 1307 m with a variability of 30 m and a mean cloud depth of around 566 m, above the observation station at 40 m a.s.l. Fig. 5a reveals a diurnal variability during the day, which affects the cloud depth. The Sc cloud is stable at night until 10:00 LT, when surface heat flux increase. This produces a reduction in cloud depth until 20:00 LT, when the CB lowers again, increasing the cloud depth.

The variability in cloud depth is partly explained by the interplay between the following processes. First, during the day, surface sensible and latent heat fluxes become dominant. Large values of the sensible heat flux combined with orographic effects lead to uplifting of the CB. This explains the diurnal variability in the CB, where heat surface fluxes are higher over the inland stations than stations close to the ocean. Furthermore, above the CT, the long wave radiative cooling lead to cloud top turbulence that entrains warm and dry air. This yields to thin the Sc cloud layer (Duynkerke et al., 1995). Unfortunately, these patterns are not identifiable from our observational data, but they are analysed via a modeling approach at the end of this section and in

section six of this manuscript. Secondly, the subsidence is the process related to the limited variability in the CT (± 30 m). Although the Pacific Anticyclone located over the research area during the summer moves NW during the winter, it is always mainly located over the SSEP, which means that small changes in seasonal subsidence produce small changes in CT. To complete the analysis and describe the relationship between Sc and our cloud-fog observations in greater depth, we describe the diurnal variability of a marine Sc cloud under well-mixed conditions. Three CB values have been estimated from the observation stations located at 40 m a.s.l. (solid line), 1090 m a.s.l. (dashed line), and 1219 m a.s.l. (dotted line) (Fig. 5b). We set the reference pressure at 40 m a.s.l. Fig. 5b shows the diurnal variability in the cloud characteristics on July 21st, 2015. Our findings provide evidence that Sc cloud is always deeper close to the ocean than further inland. We observe this throughout the day, where the cloud layer estimated over station at 40 m a.s.l. is at least hundred metres thicker than the cloud estimated at 1090 m and 1219 m. Two processes explain this spatial distribution. First, the surface heat fluxes, which produce a thinning of the marine Sc cloud when it arrives into the land. The second process is the uplifting motion of the Sc cloud when it reaches the mountains, as topography forces the cloud to rise. Our CT estimations based on the surface observations are restricted to determine which process governs this pattern. However, the modeling results shown in Fig. 6 enable us to observe reductions in the cloud depth from the ocean (solid line) to the land (solid colour), which we interpret as a thinning.

Fig. 5b shows that the altitude at which fog is observed coincides with the cloud layer inferred by observations. For example, fog observed at 1219 m a.s.l. (dots) lies within the estimated cloud layer from 1219 m a.s.l. (between the dotted line and the dashed line of CT). The latter can be interpreted as a validation of our calculations of cloud depth because the fog observations are located inside the cloud layer estimated by surface observations. During the night and early morning (00:00–10:00 LT), the Sc cloud depth remains constant and fog is only observed at the highest level (1219 m). This is closely coupled with well-mixed regimes and marine advection. On the other hand, at midday (12:00 to 18:00 LT) the cloud depth diminishes above the 40 m and 1090 m stations, and at the 1219 m station. The CB reaches the CT, resulting in dissipation of clouds. This coincides with no fog observed at 15:00 LT either, where we find a stratification of θ , and therefore a stratified regime (Figs. 3, 15:00 LT). Finally, at night (from 22:00 LT), the cloud layer grows because the heating from surface stops and fog is observed at the 1090 m and 1219 m stations, both of which were inside their estimated cloud layers (Fig. 5b).

To test the reability of our estimates of the Sc cloud characteristics retrieved from surface observation, we led the same situation using

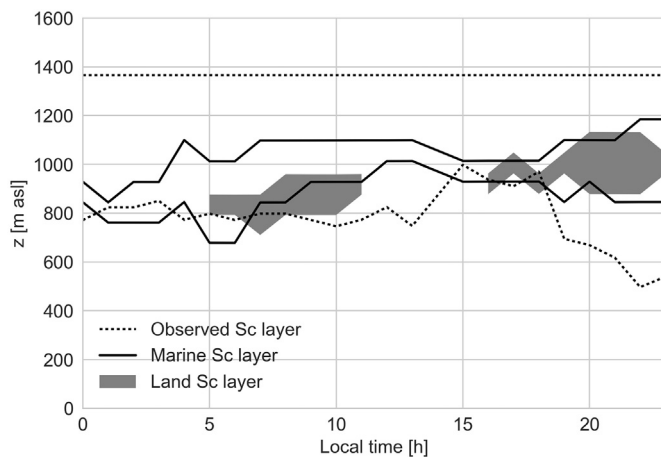


Fig. 6. Sc cloud layer modeled above a marine (solid line) and land (solid colour) WRF grid point, on July 21st, 2015. The inferred Sc cloud layer is also shown (dotted line).

WRF. Fig. 6 shows the cloud layer modeled above two points on the study area. The first is over the sea approximately 50 km from the shore (Marine Sc layer). The second one is at the site of the 40 m station (Land Sc layer). The Sc cloud layer inferred from surface observations is shown as well (dotted line).

The estimated CB agrees well with the one modeled by WRF. The modeled cloud over the land shows differences in CB of around 78 m in the night-morning (00:00 to 10:00 LT) period, with daily means of 140 m. Likewise, modeled cloud over the sea shows differences around 46 m and 70 m in the night-morning phase and daily means, respectively. The CB lies at 800 m in the morning and rises during the afternoon to around 900 m, but it diminishes differently from evening to night. The diurnal CT variability reaches its lowest level during the night and rises again during the morning, stabilising during the day and increasing again in the evening. It ranges from 900 to 1200 m, being several metres lower than the CT estimated from remote-sensing observations.

The model shows broken clouds over the land at midday and afternoon (10:00–16:00 LT), which coincide with the intermittent fog collection observed (Fig. 5b). Furthermore, over the sea the Sc cloud continues to thin from its CB and CT at midday and during the afternoon. The model confirms the diurnal thinning and break of the cloud layer over both land and sea (Figs. 3, 4 and 5b). On the one hand, we observe the presence of the well-mixed regime during the night-morning phase and the change to a stratified regime from 11:00 LT to 16:00 LT. On the other hand, we confirm the thinning process of the Sc clouds as they move from sea to land (Fig. 5b). Fig. 6 always shows a thinner cloud over the land (solid colour) than over the sea (solid line). Cereceda et al. (2008b) also observed a CB and CT between 700 m and 1200 m, but also broken clouds in the MS-LF zone in July 2002.

In summary, the diurnal variability of the marine Sc cloud is produced by a complex set of interactions between the MBL and CBL, understood as the interaction of marine Sc clouds with topography (MS-LF). The upward and downward heat fluxes lead these variabilities, resulting in changes in the cloud depth, liquid water mixing ratio (q_l) and consequently in fog observed.

5.2. Cloud liquid water mixing ratio q_l

In order to complete the characterization of the Sc cloud layer, we quantified the liquid water mixing ratio of cloud and fog. Fig. 7 shows the vertical profile of adiabatic liquid water mixing ratio (q_{la}), liquid water mixing ratio mixed with the environment (q_{lm}) and the liquid water mixing ratio obtained by WRF (q_{lw}), for different times on July 21st, 2015. The q_{la} is inferred by the parcel method without taking into account mixing with the environment during the uplifting. Thus, the q_l follows the wet adiabatic inside the cloud. The q_{lm} is converted from q_{la} by using a correction factor $m = 1$, which considers a maximum mixing with the environment during the uplifting process. Fog observations are also depicted as heights.

In the morning at around 09:00 LT, q_{lw} and q_{la} coincide in the CB, presenting LCLs at 792 m and 772 m, respectively. However, q_{lm} sets the LCL at 935 m, due to the dry air that the mixed layer incorporates into the air parcel during the rise. The morning values of q_{lw} are around 0.3 g kg^{-1} . For q_{lm} the amount of water reaches 1 g kg^{-1} at the estimated CT. The q_{la} is clearly overestimated because it does not consider the mixing during the lifting. The values reported by Duynkerke et al. (1995) are around $0.3\text{--}0.5 \text{ g kg}^{-1}$. Therefore, q_{lw} agrees better than q_{lm} and q_{la} . Finally, the amount of fog collected in L m^{-2} has the same distribution at different altitudes as q_{lm} . For example, inside the cloud layer, we observe an increase of q_l with height, and more fog is observed at the highest station (1219 m).

At 15:00 LT, on the one hand, the model estimates very low values of q_l ($< 0.3 \text{ g kg}^{-1}$), which agrees with our observations of the amount of fog (0 L m^{-2}). This drop in liquid water mixing ratio coincides with the observed breaks in the Sc cloud and the dissipation of the fog,

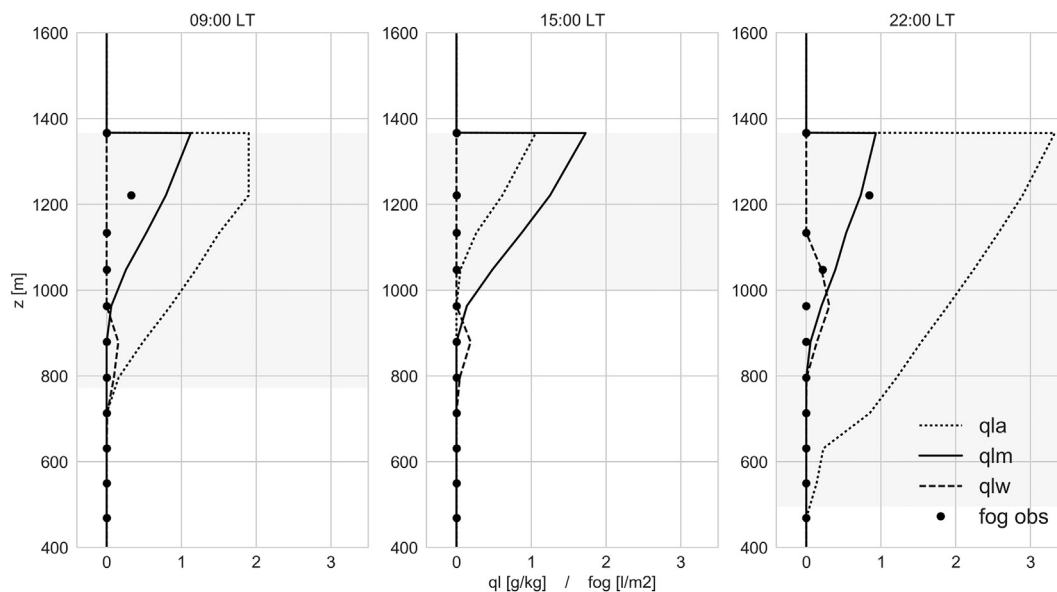


Fig. 7. Vertical profile of adiabatic liquid water mixing ratio (q_{la}), liquid water mixing ratio considering mixing with environment (q_{lm}) and liquid water mixing ratio given by WRF (q_{lw}). For morning, afternoon and night at July 21st, 2015. Fog is shown by dots and cloud layer is shown as solid colour.

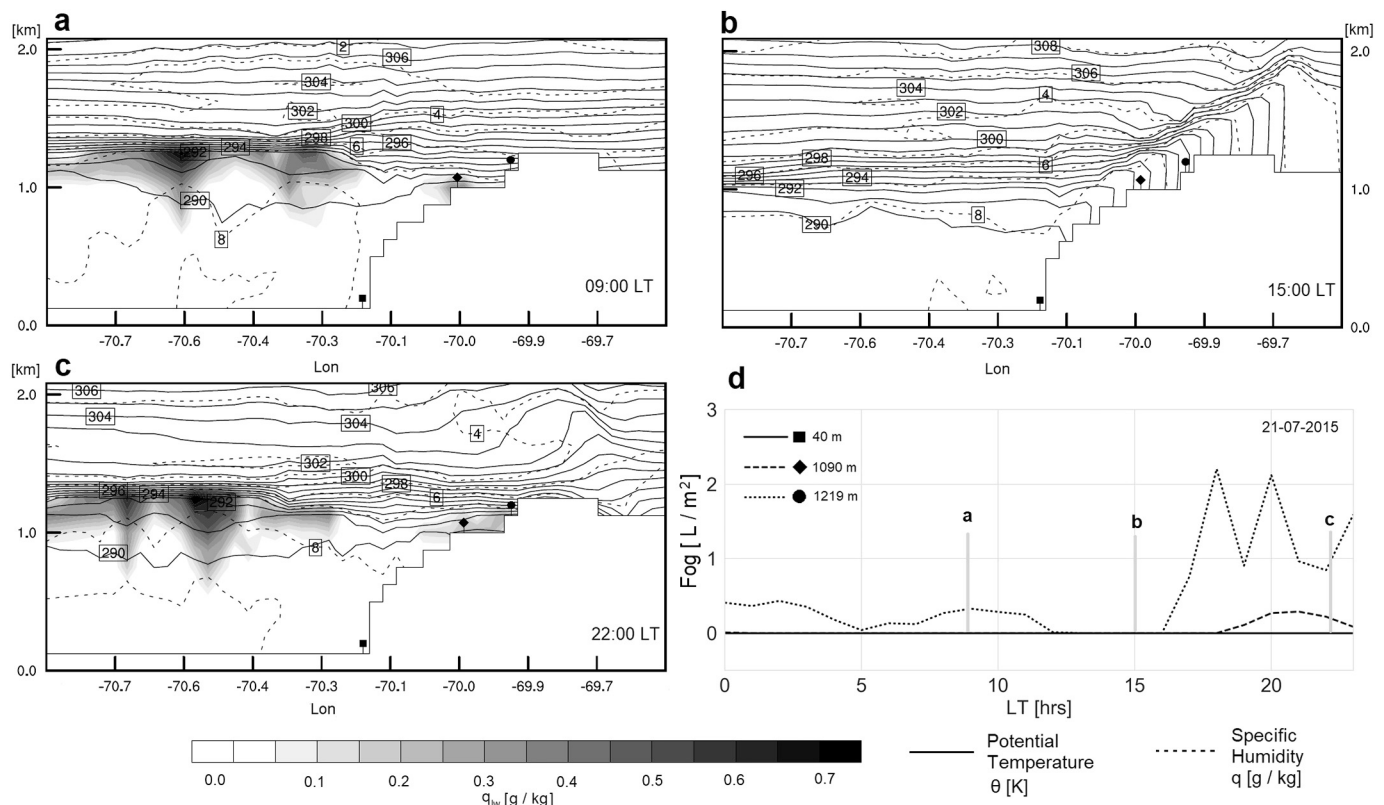


Fig. 8. (a), (b) and (c), Cross-section of liquid water mixing ratio, q_{lw} (grey scale), potential temperature θ (solid line) and specific humidity q (dashed line) by WRF on July 21st, 2015. The observation stations are shown over the digital elevation model. (d), fog observations on July 21st, 2015 at different altitudes.

related to the stratified regimes which are shown in Figs. 3, 5b and 6. On the other hand, both q_{lm} and q_{la} show values over 1 g kg^{-1} or even almost 2 g kg^{-1} . Here, q_{lm} is much higher than q_{la} , which can be explained in terms of two processes. First, the mixed layer is more humid than the surface layer (Fig. 3b). Then, during the rising of the air parcel, q increases with height. The second process is an uplifting of well-mixed parcel air, driven by the topography, where the same parcel is forced to rise by the topography while maintaining the same cloud water

conditions. However, under stratified conditions, such as those shown at 15:00 LT (Fig. 7), the parcel method gives unrealistic estimates. Therefore, we maintain our analysis of the simulated values of q_{lw} .

Finally, at 22:00 LT, q_{lw} and q_{lm} increase similarly, setting the LCLs at 878 m and 849 m, respectively. Both q_{lw} and q_{lm} coincide in the core of the cloud at around 950 m. At the same time, fog also increases with height from 0.3 L m^{-2} to 1 L m^{-2} . The amounts of water mixing ratio are around 0.4 g kg^{-1} in the model and 0.7 g kg^{-1} in q_{lm} . We observed a

reduction in q_{lw} from 1130 m, but not in q_{lm} . This is explained by the fact that the WRF model includes dry entrainment from the free troposphere, while q_{lm} does not. These increases in water mixing ratio and fog coincide with the presence of a well-mixed regime (Fig. 3).

In conclusion, the mean values of q_{lw} and q_{lm} agree with the amounts of liquid water mixing ratio reported by Duynkerke et al. (1995), Stevens et al. (2003), Garreaud and Munoz (2004) and Hogan et al. (2005), in their works on marine Sc clouds. From Fig. 7 we conclude fog collection is closely related to threshold values of q_l that Sc cloud and therefore fog contain. This is because their values increase in similar patterns than q_{lw} does. However, the values of q_l must be higher than 0.3 g kg^{-1} to be observed by the collectors. Finally, the use of the mixing factor $m = 1$ suggests a more realistic liquid water mixing ratio of Sc cloud and fog. However, other processes such as air entrainment from the free troposphere and microphysics of cloud also need to be considered.

6. Vertical and horizontal Sc cloud-fog variability

To support our analysis, we simulated the vertical MS-LF variability and its spatial distribution by means of a numerical experiment in WRF. The experiment reveals a complex interplay between the marine Sc cloud and local factors such as topography, local circulation, and synoptic conditions. These interactions enhance the mixing of the ABL by local land-sea circulation and marine Sc cloud advection, which explains the formation and maintenance of fog. The same set of interactions also produces the thermal stratification that dissipates the fog.

6.1. Vertical Sc cloud-fog variability

Fig. 8 shows a West-East cross-section of the Atacama Desert coastal zone, depicting the liquid water mixing ratio q_{lw} , potential temperature and specific humidity at 09:00, 15:00 and 22:00 LT on July 21st, 2015 (well-mixed case). The surface observations are placed above the topography and the figure also includes a diurnal distribution of fog observations (Fig. 8d). The experiment models a MBL developed from the ocean surface to 900–1250 m a.s.l. This has a well-defined structure of well-mixed θ and q conditions that reveal the presence of the turbulence that drives Sc cloud formation. The MBL is blocked by topography, which acts as a barrier that prevents marine conditions from having a significant influence farther inland. This influence reaches only as far as 20 km inland and 1250 m height (Fig. 1).

At 09:00 LT (Fig. 8a), the experiment shows the formation of a thin layer of q_{lw} , which represents the marine Sc cloud. This spans a range of altitudes from 850 to 900 m to around 1200 m. The values of q_{lw} are within the range of 0.2 to 0.5 g kg^{-1} , with maximum collections of about 0.7 g kg^{-1} . As Fig. 7 shows, q_{lw} possesses a vertical structure that increases with altitude and has its highest values of q_{lw} close to the top of the boundary layer. The experiment shows vertical moisture fluxes ($2 \cdot 10^{-3} \text{ g kg}^{-1} \text{ m}^{-1}$), lifting from the ocean to the cloud at longitude -70.6 as Fig. 8a shows. The marine Sc cloud tends to break down before colliding with the topography, sinking slightly just before arriving on land due to katabatic winds. This breakdown is accompanied by a drop in specific humidity and an increase in θ stratification, due by dry air entraining from the free troposphere at approximately -70.1 longitude and an altitude of 800 m. Here, at the top of the Sc, shortwave radiation partially offsets the longwave radiative cooling. This process, as well as the entrainment of dry air, contributes to thinning and dissipating the Sc cloud. However, the surface fluxes compensate for and maintain the cloud as shown by the well-mixed condition of θ and q shown in observations Figs. 3a and 8b (09:00 LT), but also in the simulation illustrated in Fig. 8a. Fog observations on land show amounts of only around 0.3 L m^{-2} collected at 1219 m (black circle). However, the simulation does not show clouds forming at that height in the morning, instead placing them several metres below. During the morning we find a clearly well-mixed ABL (θ and q), which coincides

with the regime illustrated in Fig. 3. The value of θ measured at 09:00 by the stations located at 1090 m and 1219 m is 292 K, which satisfactorily matches with the values derived by the experiment. The q measured by the land meteorological stations also fits well with the model, with values within the range of $7\text{--}7.5 \text{ g kg}^{-1}$.

During the afternoon, at 15:00 LT (Fig. 8b) experiment shows a stratification of θ and q between the altitudes of 800–1200 m, which promotes Sc cloud-fog dissipation. Regarding the latter value, the observations (Fig. 3) show a $\partial\bar{\theta}/\partial z$ of about $6 \cdot 10^{-3} \text{ K m}^{-1}$ (Table 1), which fits well with the experimental observations. The q evidences a strong uplift of humid air at the slope facing the ocean (anabatic winds) that emerges from the better-mixed lower layers of the ABL and the ocean. However, the thermal stratification itself remains at the slope, dissipating the Sc cloud and the fog. During the afternoon, we observe changes in cloud deck, which dissipates because of the rise of dry air entrainment and shortwave radiation at the top of the cloud. These processes enhance the thinning and dissipation of clouds mixing with the dryer air, showing a vertical stratification of θ . This stratification is present both in the observations shown in Figs. 3a at 15:00 LT, but also in the simulation (Fig. 8b). Fig. 8d does not show any observations of fog at 15:00 LT, which agrees with the experiment.

Finally, at 22:00 LT (Fig. 8c), the simulation once again reveals the development of a well-mixed ABL, which reaches a height of 1200 m. We observe the formation of a more robust cloud layer (q_{lw}) at the top of the boundary layer, which is thicker than the cloud layer that formed during the morning, but also with higher values of liquid water (0.5 to 0.7 g kg^{-1}). These values of q_{lw} satisfactorily agree well with the large amounts of fog observed at the top of the boundary layer, which reach values as high as 0.5 to 2.1 L m^{-2} . The cloud formation above the stations contains from 0.2 to 0.4 g kg^{-1} and matches the fog observations (Fig. 8). Both θ and q show a predominant well-mixed regime that extends to a height of 1250 m. Likewise, vertical upward fluxes of humid air from the ocean penetrate the cloud layer, providing evidence of the well-mixed process. At night, the breakdown of the cloud also takes place before it arrives at the mountains, but with a weaker downward flux of dry air than in the morning. Finally, the thick marine ABL influence the land above 1250 m and Eastward of longitude 70 W.

Fig. 9 shows the same West-East cross-section of Fig. 8, but for the stratified regime case on July 28th, 2015 (Fig. 4). The figure includes the surface observations above the topography and a diurnal distribution of fog observations (Fig. 9d). The experiment models a shallow MBL developed from the ocean surface to 700–900 m a.s.l. at the night-morning and to 500 m a.s.l. during the afternoon. This has a well-defined stratified structure of θ and q , conditions that reveal the presence of a stable layer that drives Sc cloud dissipation. The shallow MBL produces a null influence inland due to the topography and its shallow formation (Fig. 9). In summary, the numerical experiment agrees with our surface observations in terms of θ , q , and the relation q_{lw} -fog, for both well-mixed and stratified cases. Also, the experiment confirms the relationship between the well-mixed regime and Sc cloud-fog formation and maintenance, as well as the stratified regime with Sc cloud-fog dissipation. Our findings also corroborate our initial hypothesis regarding the influence exerted by the MBL over land during fog events (well-mixed dominant regime) and the influence of the CBL over land during fogless events (stratified conditions).

6.2. Synoptic conditions of Sc cloud-fog spatial distribution

Fig. 10 shows the spatial distribution of wind-speed and direction and cloud water mixing ratio of the MS-LF in the WRF experiment on July 21st, 2015 at 09:00, 15:00 and 22:00 LT. The figure shows a horizontal plane of the Atacama coastal desert, with the Pacific Ocean on the West and the continent on the East. The liquid water path, LWP [kg m^{-2}] of the first 150 m from the surface, is represented as fog, but also as a Sc cloud in the range 0–1500 m (greyscale). We decided to quantify Sc cloud-fog by using the LWP because it includes a sum of a q_l

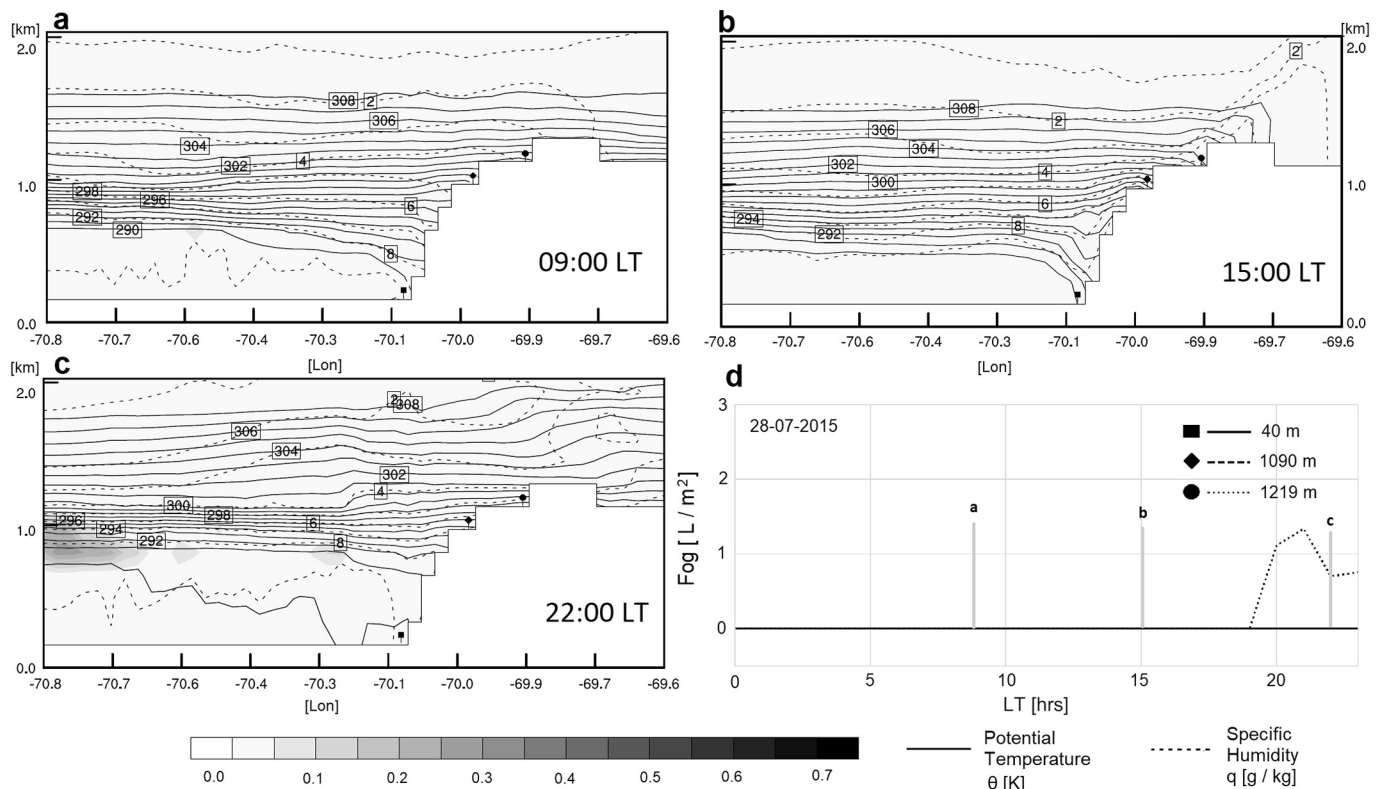


Fig. 9. (a), (b) and (c), Cross-section of liquid water mixing ratio, q_{lw} (grey scale), potential temperature θ (solid line) and specific humidity q (dashed line) by WRF on July 28th, 2015. The observation stations are shown over the digital elevation model. (d), fog observations on July 28th, 2015 at different altitudes.

vertical column, expressed as a surface value m^{-2} . The wind speed and direction also represent the regional circulation. Finally, the pressure [hPa] at surface level is shown, for both land and sea.

The topography influences the atmospheric local circulation by acting as a physical barrier to the synoptic winds that blow from S-SW. This is shown in Fig. 10 as a ridge of surface-level pressure close to the coast. Moreover, due to the diurnal land convection, pressure decrease of about 1 hPa enhances the marine advection into the land.

At about 09:00 LT, katabatic winds from the land to the ocean predominate as they descend the steep slope. At the same time, at 1500 m synoptic winds blow from the ocean to the land, which advects clouds with LWP above $50 kg m^{-2}$. The returned circulation enhances the mixing process, as Fig. 10a illustrates, and raising the ABL height (Fig. 8a). As the figure shows, patches of fog form over the land, with values around $10\text{--}15 kg m^{-2}$. However, these values do not match our surface observations (Fig. 8d), where fog is scarcely observed at 1219 m ($0.3 L m^{-2}$). Nevertheless, patches of Sc clouds form at altitudes of 150–1500 m above the 40 m station, as the estimations retrieved by our observations suggest (Fig. 5b). We conclude that well-mixed regime in the morning is produced and reinforced by the land-sea local circulation.

During the afternoon, at around 15:00 LT, surface-level pressure drops over land (1015 to 1014 hPa), increasing the advection of marine air masses from the S-SW and at speeds of around $10 m s^{-1}$. These anabatic winds face the slope and lead to an uplift of moisture, which is transported inland as Fig. 8b shows, while θ remains stratified. At 1500 m, winds with similar speeds, but from the west are blocked by the mountain barrier. Thermal stratification and moisture advection can also be seen in our observations (Fig. 3), where at around 15:00 LT, θ displays larger differences than q . We conclude that this process explains Sc cloud-fog dissipation. In validation of our findings, no patches of fog are shown either by the model at the stations nor in our observations, although some Sc cloud patches were observed over the sea

and below 1500 m.

At night, around 22:00 LT, the surface level pressure rises by around 2 hPa, due to the cooling effect over land and to subsidence (Fig. 10c), partially inhibiting the inland advection of the marine air masses. Therefore, winds at surface level are deflected from SW to the NW due to topography barrier. The circulation at 1500 m (843 hPa), is still characterized by SW winds, which pass over the mountain, although their speed drops from 10 to $5 m s^{-1}$. The θ and q profiles are well-mixed, due to a combination of effects: (a) local returned circulation, (b) marine Sc cloud advection over land and (c) pressure rise. The mixing produces fog at several places around the mountains, which agree with fog observations (Fig. 8d). The model is capable of predicting the presence of fog at stations 1219 m (black circle) and 1090 m (black diamond) with values in the range $15\text{--}40 kg m^{-2}$. At 1500 m, we find a strong advection of marine Sc cloud from SW, with values above $50 kg m^{-2}$, which also agrees with observations in which Sc clouds form above all the stations (Fig. 5b). The Sc cloud area covered by the numerical experiment agrees with reported Sc cloud satellite observations made by Cereceda et al. (2008b) for July 2002.

In summary, we confirm that fog is produced and maintained under a well-mixed regime, in which local circulation, topography, and subsidence all play key roles. This phenomenon occurs in the morning and evening-night (09:00 and 22:00 LT), when advection of marine Sc air masses facilitates fog formation. During the afternoon (15:00 LT), changes in synoptic conditions and therefore local circulation produce a thermal stratification that dissipates clouds.

7. Conclusions

We studied the interaction between marine Sc clouds and land fog by combining surface observations and high-resolution numerical experiments in the Weather Research and Forecasting Model (WRF). Our findings show that this interaction involves either a well-mixed

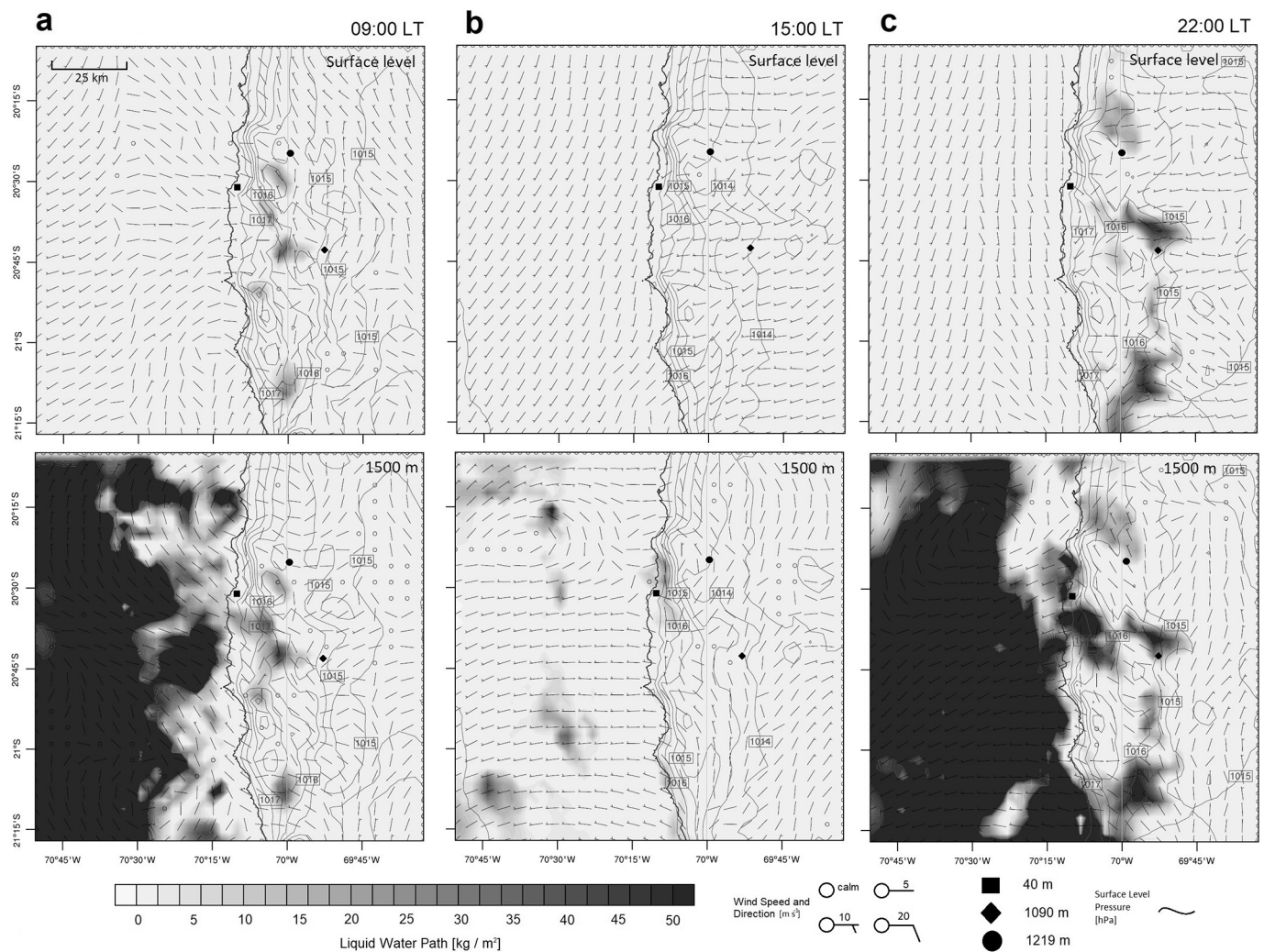


Fig. 10. Sc cloud-fog, wind speed-direction and surface-level pressure spatial distribution according to the WRF model on July 21st, 2015. Sc cloud-fog water mixing ratio is represented as liquid water path, LWP (grey scale), for the first 150 m and the first 1500 m. Both variables are plotted at 09:00 LT (a), 15:00 LT (b) and 22:00 LT (c). Meteorological stations are also shown.

condition or a thermal stratification on the ABL, which forms or dissipates Sc cloud-fog.

The well-mixed regime is characterized by low gradients of potential temperature ($< 3.10 \cdot 10^{-3} \text{ K m}^{-1}$) and specific humidity ($< 1.60 \cdot 10^{-3} \text{ g kg}^{-1} \text{ m}^{-1}$). These well-mixed conditions, which are driven mainly by the marine Sc cloud deck, have a small diurnal variability. During the morning hours, the influence of marine Sc cloud advection and the katabatic winds that fall from the slopes enhance the well-mixed condition near the coast. Around midday, increases in surface heat fluxes lead to larger gradients in potential temperature, but enhance the moisture being uplifted through the slope. At night, synoptic winds and Sc cloud advection are blocked by the topography and inversion capping, resulting in a returned local circulation, which reinforces the well-mixed condition. The predominance of the well-mixed regime correlates with fog observations, which are measured in the morning and evening-night.

In contrast, the stratified regime is characterized by high gradients of potential temperature ($> 3.10 \cdot 10^{-3} \text{ K m}^{-1}$) and specific humidity ($> 1.60 \cdot 10^{-3} \text{ g kg}^{-1} \text{ m}^{-1}$). This regime is affected by the diurnal variability, which displays the largest changes on diurnal gradients around midday. In the morning of days dominated by stratified regimes, the gradients in the upper levels of the ABL are low often, which produces fog. However, around midday, the surface heat fluxes increase inland, resulting in an uplifting of small amounts of moisture (lower

than 0.3 g kg^{-1}), but also a strong stratification of potential temperature (differences until 19 K). The thermal stratification corresponds with fogless observations, produced mainly between 12:00 and 20:00 LT. We can therefore conclude that the stratified regime leads to optimal conditions for the dissipation of Sc cloud-fog. Our results confirm the usefulness of surface observations analysed by means of a simple parcel method to obtain a first estimate of cloud characteristics such as cloud depth, liquid water mixing ratio and their association with fog over land. Together with the analysis of WRF results, this has enabled us to provide an explanation of the role played by marine Sc cloud in fog formation, and of the land processes involved.

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.atmosres.2018.07.009>.

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